

Tutorial 5 – Factor Models

MMF 1921H Operations Research

Prepared by: Giorgio Costa¹

¹gcosta@mie.utoronto.ca

Consider a generic multi-factor model with p factors and n assets. The return of asset i can be explained by the factors as follows

$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}.$$

Consider a generic multi-factor model with p factors and n assets. The return of asset i can be explained by the factors as follows

$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}.$$

Given some data (e.g., historical observations), we may have

- ▶ Data on asset i for $i = 1, \dots, n$.
- ▶ Data on factor k for $k = 1, \dots, p$.
- ▶ Individual observations t for $t = 1, \dots, T$.

$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}.$$

Let's say we have the following data for some asset i :

1. Observed returns of asset i over time $t = 1, \dots, T$: $\mathbf{r}_i \in \mathbb{R}^T$

$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}.$$

Let's say we have the following data for some asset i :

1. Observed returns of asset i over time $t = 1, \dots, T$: $\mathbf{r}_i \in \mathbb{R}^T$
2. Observed factor returns over time $t = 1, \dots, T$: $\mathbf{f} = [\mathbf{f}_1 \ \mathbf{f}_2 \ \dots \ \mathbf{f}_p] \in \mathbb{R}^{T \times p}$
 - Note: Each column corresponds to a single factor over time $t = 1, \dots, T$

$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}.$$

Let's say we have the following data for some asset i :

1. Observed returns of asset i over time $t = 1, \dots, T$: $\mathbf{r}_i \in \mathbb{R}^T$
2. Observed factor returns over time $t = 1, \dots, T$: $\mathbf{f} = [\mathbf{f}_1 \ \mathbf{f}_2 \ \dots \ \mathbf{f}_p] \in \mathbb{R}^{T \times p}$
 - Note: Each column corresponds to a single factor over time $t = 1, \dots, T$
3. Data matrix: $\mathbf{X} = [\mathbf{1} \ \mathbf{f}] \in \mathbb{R}^{T \times (p+1)}$

Note: We have included a column of ones to account for the intercept of regression ('alpha')

$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}.$$

Let's say we have the following data for some asset i :

3. Data matrix: $\mathbf{X} = [\mathbf{1} \ \mathbf{f}] \in \mathbb{R}^{T \times (p+1)}$

Note: We have included a column of ones to account for the intercept of regression ('alpha')

4. Vector of regression coefficients: $\mathbf{B}_i = \begin{bmatrix} \alpha_i \\ \mathbf{V}_i \end{bmatrix} = \begin{bmatrix} \alpha_i \\ \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{ip} \end{bmatrix} \in \mathbb{R}^{p+1}$

- Note: The vector $\mathbf{V}_i$ represents our factor loadings (or 'betas')

$$\mathbf{V}_i = \begin{bmatrix} \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{ip} \end{bmatrix} \in \mathbb{R}^p$$

$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}.$$

Let's say we have the following data for some asset i :

4. Vector of regression coefficients: $\mathbf{B}_i = \begin{bmatrix} \alpha_i \\ \mathbf{V}_i \end{bmatrix} = \begin{bmatrix} \alpha_i \\ \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{ip} \end{bmatrix} \in \mathbb{R}^{p+1}$

► Note: The vector $\mathbf{V}_i$ represents our factor loadings (or 'betas')

$$\mathbf{V}_i = \begin{bmatrix} \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{ip} \end{bmatrix} \in \mathbb{R}^p$$

5. Vector of residuals for asset i over time $t = 1, \dots, T$: $\varepsilon_i \in \mathbb{R}^T$

For ordinary least squares (OLS) regression, we wish to minimize the sum of squared residuals. This is an unconstrained minimization problem,

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2$$

where $\|\cdot\|_2$ is the ℓ_2 norm operator.

For ordinary least squares (OLS) regression, we wish to minimize the sum of squared residuals. This is an unconstrained minimization problem,

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2$$

where $\|\cdot\|_2$ is the ℓ_2 norm operator.

Using basic calculus, we can find a closed-form solution to the OLS model, where

$$\Rightarrow \mathbf{B}_i^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{r}_i$$

If we want to calibrate n OLS models (for each asset), we can extend our closed-form solution as follows

$$\mathbf{B} = \begin{bmatrix} \boldsymbol{\alpha}^T \\ \mathbf{V} \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}$$

If we want to calibrate n OLS models (for each asset), we can extend our closed-form solution as follows

$$\mathbf{B} = \begin{bmatrix} \boldsymbol{\alpha}^T \\ \mathbf{V} \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R}$$

- ▶ Matrix of asset returns : $\mathbf{R} \in \mathbb{R}^{T \times n}$

Note: Each column of $\mathbf{R}$ corresponds to all the observations for $t = 1, \dots, T$ for each individual asset.

- ▶ Vector of intercepts for all n assets: $\boldsymbol{\alpha} \in \mathbb{R}^n$

- ▶ Matrix of factor loadings for all n assets: $\mathbf{V} \in \mathbb{R}^{p \times n}$

Note: Each column of the matrix $\mathbf{V}$ corresponds to the factor loadings for each asset i ,

$$\mathbf{V}_i = \begin{bmatrix} \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{ip} \end{bmatrix} \in \mathbb{R}^p$$

The residuals resulting from the optimal regression coefficients are

$$\varepsilon_i = \mathbf{r}_i - \mathbf{X}\mathbf{B}_i^*$$

The residuals resulting from the optimal regression coefficients are

$$\varepsilon_i = \mathbf{r}_i - \mathbf{X}\mathbf{B}_i^*$$

We can use this vector of residuals to calculate the unbiased estimate of the residual variance

$$\sigma_{\varepsilon_i}^2 = \frac{1}{T - p - 1} \|\varepsilon_i\|_2^2$$

where we divide by the appropriate number of degrees of freedom (DOF), $T - p - 1$. This is because we have T observations, but we use these observations to calculate $p + 1$ coefficients (one 'alpha' and p 'betas').

Estimate MVO input parameters:

Scalar notation

Asset returns $r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}$

Exp. returns $\mu_i = \alpha_i + \sum_{k=1}^p \beta_{ik} \bar{f}_k$

Var./Cov. $\sigma_i^2 = \sum_{k=1}^p \beta_{ik}^2 \sigma_{f_k}^2 + \sigma_{\varepsilon_i}^2$

$$\sigma_{ij} = \sum_{k=1}^p \beta_{ik} \beta_{jk} \sigma_{f_k}^2$$

Estimate MVO input parameters:

Scalar notation

Asset returns
$$r_{it} = \alpha_i + \sum_{k=1}^p \beta_{ik} f_{kt} + \varepsilon_{it}$$

Exp. returns
$$\mu_i = \alpha_i + \sum_{k=1}^p \beta_{ik} \bar{f}_k$$

Var./Cov.
$$\sigma_i^2 = \sum_{k=1}^p \beta_{ik}^2 \sigma_{f_k}^2 + \sigma_{\varepsilon_i}^2$$

$$\sigma_{ij} = \sum_{k=1}^p \beta_{ik} \beta_{jk} \sigma_{f_k}^2$$

Vector notation

$$\mathbf{r}_t = \boldsymbol{\alpha} + \mathbf{V}^T \mathbf{f}_t + \boldsymbol{\varepsilon}_t$$

$$\boldsymbol{\mu} = \boldsymbol{\alpha} + \mathbf{V}^T \bar{\mathbf{f}}$$

$$\mathbf{Q} = \mathbf{V}^T \mathbf{F} \mathbf{V} + \mathbf{D}$$

Estimate MVO input parameters:

Vector notation

Asset returns $\mathbf{r}_t = \alpha + \mathbf{V}^T \mathbf{f}_t + \varepsilon_t$

Exp. returns $\mu = \alpha + \mathbf{V}^T \bar{\mathbf{f}}$

Var./Cov. $\mathbf{Q} = \mathbf{V}^T \mathbf{F} \mathbf{V} + \mathbf{D}$

where

- ▶ Vector of random asset returns at time t : $\mathbf{r}_t \in \mathbb{R}^n$
- ▶ Vector of random factor returns at time t : $\mathbf{f}_t \in \mathbb{R}^p$
- ▶ Vector of expected factor returns: $\bar{\mathbf{f}} \in \mathbb{R}^p$
- ▶ Factor covariance matrix: $\mathbf{F} \in \mathbb{R}^{p \times p}$
- ▶ Diagonal matrix of residual variances: $\mathbf{D} \in \mathbb{R}^{n \times n}$

Note: The diagonal elements of this matrix are the residual variances, $\sigma_{\varepsilon_i}^2$.
All off-diagonal elements are zero.

Correlated factors:

If the factors are correlated, we must include this effect into our calculation of the asset variance and covariance. In other words, if $\text{cov}(f_i, f_j) \neq 0$, the factor covariance terms $\sigma_{f_i f_j}$ will manifest in the resulting asset variance/covariance. If we use the expression

$$\mathbf{Q} = \mathbf{V}^T \mathbf{FV} + \mathbf{D}$$

to calculate our variances and covariances, then this will automatically embed all the appropriate factor covariance terms.

Three norms

- ▶ ℓ_0 norm:

$$\|\mathbf{B}_i\|_0 = \mathbb{1}_{\alpha_i \neq 0} + \sum_{k=1}^p \mathbb{1}_{\beta_{ik} \neq 0}$$

Note: ' $\mathbb{1}_{\beta_{ik} \neq 0}$ ' denotes the indicator function on whether the condition $\beta_{ik} \neq 0$ is satisfied.

Three norms

- ▶ ℓ_0 norm:

$$\|\mathbf{B}_i\|_0 = \mathbb{1}_{\alpha_i \neq 0} + \sum_{k=1}^p \mathbb{1}_{\beta_{ik} \neq 0}$$

Note: ' $\mathbb{1}_{\beta_{ik} \neq 0}$ ' denotes the indicator function on whether the condition $\beta_{ik} \neq 0$ is satisfied.

- ▶ ℓ_1 norm:

$$\|\mathbf{B}_i\|_1 = |\alpha_i| + \sum_{k=1}^p |\beta_{ik}|$$

Note: The ℓ_1 norm is the sum of absolute values of all elements of the vector $\mathbf{B}_i \in \mathbb{R}^p$.

Three norms

- ▶ ℓ_0 norm:

$$\|\mathbf{B}_i\|_0 = \mathbb{1}_{\alpha_i \neq 0} + \sum_{k=1}^p \mathbb{1}_{\beta_{ik} \neq 0}$$

Note: ' $\mathbb{1}_{\beta_{ik} \neq 0}$ ' denotes the indicator function on whether the condition $\beta_{ik} \neq 0$ is satisfied.

- ▶ ℓ_1 norm:

$$\|\mathbf{B}_i\|_1 = |\alpha_i| + \sum_{k=1}^p |\beta_{ik}|$$

Note: The ℓ_1 norm is the sum of absolute values of all elements of the vector $\mathbf{B}_i \in \mathbb{R}^p$.

- ▶ ℓ_2 norm:

$$\|\mathbf{B}_i\|_2 = \left((\alpha_i)^2 + \sum_{k=1}^p (\beta_{ik})^2 \right)^{1/2}$$

Note: The ℓ_2 norm is also known as the Euclidean norm.

Three regression models: Constrained form

$$\begin{aligned} \text{Best Subset Selection:} \quad & \min_{\mathbf{B}_i} \quad \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 \\ & \text{s.t.} \quad \|\mathbf{B}_i\|_0 \leq K \end{aligned}$$

$$\begin{aligned} \text{LASSO:} \quad & \min_{\mathbf{B}_i} \quad \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 \\ & \text{s.t.} \quad \|\mathbf{B}_i\|_1 \leq s \end{aligned}$$

$$\begin{aligned} \text{Ridge Regression:} \quad & \min_{\mathbf{B}_i} \quad \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 \\ & \text{s.t.} \quad \|\mathbf{B}_i\|_2^2 \leq s \end{aligned}$$

Note 1: The parameter $K \in \mathbb{Z}_+$ in the Best Subset Selection model is a positive integer corresponding to a cardinality value. On the other hand, the positive parameter $s \in \mathbb{R}_+$ limits the size of the sum of our regression coefficients in both LASSO and Ridge Regression.

Note 2: The regularizer for Ridge Regression is the square of the ℓ_2 norm for mathematical simplicity. This allows us easily derive a closed-form solution.

Three regression models: Penalized form

$$\text{Best Subset Selection: } \min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_0$$

$$\text{LASSO: } \min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_1$$

$$\text{Ridge Regression: } \min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_2^2$$

Note 1: In the case of the Best Subset Selection model, the penalized formulation does not give us explicit control over the number of non-zero elements, i.e., we cannot explicitly state our cardinality.

Note 2: The LASSO model in its constrained form and its penalized form are equivalent. In other words, there exists some value of $s \geq 0$ and some value of $\lambda \geq 0$ such that the optimal solution $\mathbf{B}_i^*$ is the same for both versions of LASSO. The same conclusion holds for Ridge Regression.

Best Subset Selection

The ℓ_0 norm is simply trying to impose a cardinality limit on our coefficients. Thus, we can address this the same way we addressed cardinality constrained portfolios.

Best Subset Selection

The ℓ_0 norm is simply trying to impose a cardinality limit on our coefficients. Thus, we can address this the same way we addressed cardinality constrained portfolios.

Consider the constrained form of the Best Subset Selection model.

$$\begin{aligned} \min_{\mathbf{B}_i, \mathbf{y}} \quad & \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 \\ \text{s.t.} \quad & \mathbf{L}\mathbf{y} \leq \mathbf{B}_i \leq \mathbf{U}\mathbf{y} \\ & \mathbf{1}^T \mathbf{y} \leq K \\ & y_j \in \{0, 1\}, j = 1, \dots, p + 1 \end{aligned}$$

where $y_j \in \{0, 1\}$ for $j = 1, \dots, p + 1$ are auxiliary binary variables.

Best Subset Selection

The ℓ_0 norm is simply trying to impose a cardinality limit on our coefficients. Thus, we can address this the same way we addressed cardinality constrained portfolios.

Consider the constrained form of the Best Subset Selection model.

$$\begin{aligned} \min_{\mathbf{B}_i, \mathbf{y}} \quad & \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 \\ \text{s.t.} \quad & \mathbf{L}\mathbf{y} \leq \mathbf{B}_i \leq \mathbf{U}\mathbf{y} \\ & \mathbf{1}^T \mathbf{y} \leq K \\ & y_j \in \{0, 1\}, j = 1, \dots, p + 1 \end{aligned}$$

where $y_j \in \{0, 1\}$ for $j = 1, \dots, p + 1$ are auxiliary binary variables.

Note: We must select appropriate lower and upper bounds, $\mathbf{L}$ and $\mathbf{U}$. Having a large feasible region is sometimes detrimental for computational performance. On the other hand, if we set $\mathbf{L}$ and $\mathbf{U}$ to be very small in magnitude, we might shrink the feasible region and we may not be able to reach the true optimal solution.

LASSO

The ℓ_1 norm is continuous and convex, but it is not smooth everywhere.

LASSO

The ℓ_1 norm is continuous and convex, but it is not smooth everywhere.

Consider the penalized form of the LASSO model, and introduce the auxiliary variable $\mathbf{y} \in \mathbb{R}^{p+1}$ to reformulate the ℓ_1 norm term into $2p + 2$ constraints.

$$\begin{aligned} \min_{\mathbf{B}_i, \mathbf{y}} \quad & \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \mathbf{1}^T \mathbf{y} \\ \text{s.t.} \quad & \mathbf{y} \geq \mathbf{B}_i \\ & \mathbf{y} \geq -\mathbf{B}_i \end{aligned}$$

LASSO

The ℓ_1 norm is continuous and convex, but it is not smooth everywhere.

Consider the penalized form of the LASSO model, and introduce the auxiliary variable $\mathbf{y} \in \mathbb{R}^{p+1}$ to reformulate the ℓ_1 norm term into $2p + 2$ constraints.

$$\begin{aligned} \min_{\mathbf{B}_i, \mathbf{y}} \quad & \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \mathbf{1}^T \mathbf{y} \\ \text{s.t.} \quad & \mathbf{y} \geq \mathbf{B}_i \\ & \mathbf{y} \geq -\mathbf{B}_i \end{aligned}$$

Note: At optimality, the auxiliary variable $\mathbf{y}$ will be equal to the absolute value of the coefficients, i.e., $\mathbf{y} = |\mathbf{B}_i|$.